

WALLET VIBE — PLAY STORE (.AAB) & MULTIPLATFORM DEPLOYMENT INSTRUCTIONS

Mandatory Specifications for Google Play App Bundles (.aab) & Multi-Layer Web/App Synchronization

1. Google Play Store Mandatory Requirements (.aab)

- **Format Type: Official .aab (Android App Bundle):** Google Play requires all new apps and updates to be submitted as an .aab package. APKs are obsolete for Play Store publishing.
- **How It Works:** The .aab bundle contains compiled web assets, Android manifest, and native wrappers. Google Play automatically generates device-optimized APKs for each user's specific CPU and screen density.
- **Play App Signing:** App signing is managed by Google Play. Developers sign the .aab using an Upload Key, while Google signs the generated APKs delivered to users.
- **Target API Level:** Must target Android 14 (API level 34 or higher).

2. Step-by-Step TWA / Bubblewrap (.aab) Build Guide

Step 1: Install Bubblewrap CLI:

```
npm install -g @bubblewrap/cli
```

Step 2: Initialize TWA Project from Netlify WebApp:

```
bubblewrap init --manifest=https://walletvibe.netlify.app/manifest.json
```

Step 3: Digital Asset Links Verification (Hides URL Bar):

Extract SHA-256 fingerprint from upload key: `keytool -list -v -keystore android.keystore`.

Create `public/.well-known/assetlinks.json` with statement target package name and SHA-256 fingerprint so Chrome runs the app in 100% fullscreen native app mode.

Step 4: Build Official Signed .aab Package:

```
bubblewrap build
```

This outputs `app-release-signed.aab` ready for direct upload to Google Play Console.

3. Multi-Layer Synchronization Protocol for Updates

Target Layer	Update Trigger & Mechanism	Synchronization Requirement
Layer 1: Web App	Code changes pushed to GitHub main branch.	Netlify automatically triggers continuous integration build, deploying static assets to CDN edge within ~30s.
Layer 2: Browser PWA	User accesses website via browser or home screen icon.	Service Worker detects new assets, invalidates local cache version, and prompts instant update reload.
Layer 3: Play Store (.aab)	Version bump in build.gradle (e.g. <code>versionCode 102</code>).	Re-run bubblewrap build, upload updated .aab to Google Play Console Production track.
Layer 4: Cross-Device UI	Responsive CSS viewport testing across viewports.	Enforce layout parity for Mobile (360-480px), Tablet (768-1024px), Desktop (1280px+), Foldables, and ChromeOS.

4. Mandatory Quality & Parity Rules for Developers & AI Agents

1. **Zero Platform Discrepancy:** Features added to the WebApp MUST work identically in the standalone PWA and Play Store TWA .aab build.

2. **No Browser Popups:** Never use native `alert()` or `confirm()`. Always use portal-rendered custom dialogs (`CustomDialogModal.jsx`).
3. **Deterministic Duplicates:** Duplicate checks MUST enforce 12-digit RRN uniqueness, exact amounts, and merchant token matching.
4. **Offline Persistence:** Always write data changes to `localCache` snapshot alongside Firestore `writeBatch` operations.